

JEE MAIN REVISION MASTER: POLYMERS 2026 0 QUESTIONS 0% WEIGHTAGE

Subject: Chemistry | Target Scope: JEE Main Previous Years | Empirical Base: 15 Analyzed PYQs

1. High-Yield Concept Priority Matrix

Priority Tier	Concept Name	Frequency	Category
Moderate	Polymers — Core Principles & Governing Laws	7 PYQs	Core Principle
Moderate	Polymers — Applied Cases, Reactions & Traps	5 PYQs	Applied Topics
Moderate	Polymers — Standard Calculations & Problem Archetypes	4 PYQs	Quantitative Analysis

2. Master Formula Reference Card

Concept / Topic	Mathematical Formulation & Governing Equations
Polymers — Core Principles & Governing Laws	$(CH_3)_3SiCl$
Polymers — Applied Cases, Reactions & Traps	$(C_2H_5)_3Al$, $[H_2N-(CH_2+\frac{1}{6}NH_2)]$, $TiCl_4$

3. Deep-Dive Concepts & Linked PYQ Evidence

3.1 Polymers — Core Principles & Governing Laws (Moderate Yield — 7 PYQs)

Core Principle: Fundamental definitions, core postulates, and governing physical/chemical relations in Polymers.

■ **Caution / JEE Pitfall:** Carefully verify units, standard reference states, and sign conventions in Polymers.

■ **Speed Trick:** Focus on high-yield formulas and standard limiting approximations tested in Polymers PYQs.

Representative Source PYQs:

- [2024 Shift] Q#1 (MCQ): "Which of the following set of polymers are used as fibre? (i) Teflon (ii) Starch (iii) Terylene (iv) Orlon..." [Ans: N/A]
- [2023 Shift] Q#3 (MCQ): "Which of the following monomers form biodegradable polymers ?..." [Ans: N/A]
- [2020 Shift] Q#3 (MCQ): "Which of the following monomers can undergo condensation polymerisation?..." [Ans: N/A]

3.2 Polymers — Applied Cases, Reactions & Traps (Moderate Yield — 5 PYQs)

Core Principle: Special cases, experimental observations, and conceptual traps designed to test deep understanding of Polymers.

■ **Caution / JEE Pitfall:** Read questions carefully for keywords like 'incorrect', 'not true', or non-standard conditions.

■ **Speed Trick:** Elimination of extreme or unphysical answer choices improves accuracy in assertion-reason and multiple-statement questions.

Representative Source PYQs:

- [2024 Shift] Q#2 (MCQ): "The biodegradable polymer obtained by polymerisation of glycine and aminocaproic acid is..." [Ans: N/A]
- [2020 Shift] Q#2 (MCQ): "Which of the following polymer has strongest intermolecular forces of attraction?" [Ans: N/A]
- [2018 Shift] Q#1 (MCQ): "Ziegler-Natta catalyst is used to prepare..." [Ans: N/A]

3.3 Polymers — Standard Calculations & Problem Archetypes (Moderate Yield — 4 PYQs)

Core Principle: Quantitative numerical problem-solving techniques and standard question patterns frequently seen in Polymers.

■ **Caution / JEE Pitfall:** Watch for algebraic rounding, stoichiometric balancing, and boundary condition checks.

■ **Speed Trick:** Dimensional consistency checks and ratio comparisons can quickly eliminate incorrect options.

Representative Source PYQs:

- [2023 Shift] Q#4 (MCQ): "Match the List-I with List - I in the following. List-I List-II 1. Caprolactam a. 2. Vinyl chloride b. 3. Styrene c. 4. Propene d..." [Ans: N/A]
- [2022 Shift] Q#1 (MCQ): "Which of the following is correctly matched?" [Ans: N/A]
- [2019 Shift] Q#1 (MCQ): "Cis-1, 4-polyisoprene is called..." [Ans: N/A]

4. Benchmark Illustrative PYQs with Master Solutions

Benchmark Problem 1: KCET 2024 (Q#1)

Which of the following set of polymers are used as fibre? (i) Teflon (ii) Starch (iii) Terylene (iv) Orlon

- (A) (i) and (ii)
- (B) (ii) and (iii)
- (C) (iii) and (iv)
- (D) (i) and (iv)

Correct Answer: None

Solution Analysis: The polymers that are used as fibers are : Terylene : Also known as polyester, this is a synthetic polymer that is widely used as a fiber in textiles due to its durability, resistance to stretching and shrinking, and quick-drying properties. Orlon : This is a trade name for polyacrylonitrile, a synthetic polymer that is used to make acrylic fibers, known for their wool-like feel and used for sweaters, hats, and other garments. Therefore, the correct option is : Option C : (iii) and (iv)

Benchmark Problem 2: KCET 2024 (Q#2)

The biodegradable polymer obtained by polymerisation of glycine and aminocaproic acid is

- (A) nylon 6
- (B) PHBV
- (C) nylon-2, nylon-6
- (D) nylon-6, 10

Correct Answer: None

Solution Analysis: The biodegradable polymer Nylon-2, nylon 6, is an alternating polyamide copolymer of glycine $[H_2N-CH_2-COOH]$ and amino caproic acid $[H_2N(CH_2)_5COOH]$.

Benchmark Problem 3: KCET 2022 (Q#1)

Which of the following is correctly matched?

- (A) Teflon - Caprolactum
- (B) Bakelite - Novolac
- (C) Polyester - Tetrafluoroethene
- (D) Nylon - Acrylonitrile

Correct Answer: None

Solution Analysis: The correct pair is bakelite and novolac. It is formed by the polymerisation of phenol and formaldehyde monomer. While teflon is the polymer of tetrafluoroethylene, polyester is the polymer of ethyleneglycol and terephthalic acid and nylon is polymer of caprolactum.